5 (a) Two-Stage Write Filtering & Shared Validation in H

Write Event & 2-Stage Filter

Batch: Δitem, $V(\text{item}): 0 \longrightarrow 1$

1. Table stamp `date_dim` ($V = 0$) untouched

→ **Skip checks; 3 defs reuse** `Pass`

2. Batch ΔT Probe $\Delta\text{item} \bowtie \text{item}_{\text{old}}$

→ **Dup** `id=17` **found (no full scan)**

Shared Outcomes H

key: $(c, v(\text{dep}(c)))$

Condition c	Stamp	Outcome & Scope
`date_dim` key unique	`d:0`	**Pass (reused by 3 defs)**
Premise P_{R3} (contig.)	`d:1`	**Fail (hits C2 edge R3)**
`item` key unique	`i:1`	**Fail (`id=17` dup in Δ)**

6 (b) Dual Maintenance: Revoke Broken C2 Premises & Repair Definitions

Branch 1: C2 Equivalence Premise P_{R3} Fails

1. **Suspend edge** $\text{ER} \equiv_{R3} \text{ER_r1}$ **in graph** G
2. Roll back rewrite `ER_r1` → `ER` (`date_dim` join)
3. Templates Φ routed via edge revert to tentative

✓ **Prevents stale rewrite from serving wrong dates**

Branch 2: Definition Condition $C(\text{ER})$ Fails

1. **Single-edit repair: add filter** `i_is_current='Y'`
2. Re-check $C(\text{ER}^{r_2})$ at $v = 1$ → Pass → publish r_2
3. **Probe domain shift / >1 edit** → `Pending` + **diff**

✓ **Preserves core grain/measure; blocks fan-out**

4 (c) Snapshot-Bound Execution: Check & Query Share Snapshot D_s

1. `REPEATABLE READ` s

Read stamp $v_s(\text{item}) = 1$

→

2. Verify via $H_{\text{tx}}(c, v_s)$

ER r_2 valid at $v_s = 1$ (reuse)

→

3. Run $q(\text{ER}^{r_2}, p)$ **on** D_s

Return `$142.8k` & `COMMIT`

✓ **Concurrent commit ($v = 2$) during step 2/3 cannot cause q to read unverified rows or stale $v = 0$ verdicts.**